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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001
M.Sc Degree Examination – APRIL 2021

PPH806T – NUCLEAR AND PARTICLE PHYSICS

Time : 3 hrs **Max Marks :75**

SECTION – A (10X2=20)

Answer **ALL** Questions

1. What are magic numbers? Explain.
2. Define Bohr magneton.
3. What is quadrupole moment?
4. List the properties of nuclear force.
5. What are fertile materials?
6. Define scattering cross section and give its unit.
7. What is half life?
8. Explain isomerism.
9. What are leptons?
10. Define strangeness.

SECTION – B (5X5=25)

Answer **ALL** Questions

11. a) Outline the Bohr-Wheeler theory of fission.

(or)

b) Explain angular momentum and parity of a nucleus.

12. a) Establish the existence of tensor forces.

(or)

b) Discuss the Meson theory of nuclear force.

13. a) Classify the different types of nuclear reactions.

(or)

b) Write a note on the thermonuclear reactions.

14. a) Describe the phenomenon of internal conversion.

(or)

b) Write a note on gamma decay.

15. a) Discuss the types of interactions.

(or)

b) What are quarks? Discuss their flavours, charge and spin.

SECTION –C (3X10=30)

Answer any **THREE** Questions

16. Discuss in detail the salient features of the collective model of the nucleus.

17. Explain the importance of nucleon-nucleon scattering studies in the understanding of nuclear forces.

18. a) Explain the significance of Q value of a reaction. **(5+5)**
b) Write a note on the fission reactor.

19. Discuss the Fermi's theory of beta decay and explain the shape of the beta spectrum.

20. Explain in detail the classification of elementary particles.

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